

The Architectural Factorization Principle

A Graph-Theoretical Reconstruction of Feynman Diagrammatics from Green-Function Architectures

Borros Arneth, Philipps University Marburg, Justus Liebig Giessen, Germany,
borros.arneth@staff.uni-marburg.de

Abstract

Green functions and Feynman diagrams constitute two of the most successful mathematical frameworks in modern theoretical physics. Green functions describe propagation and response, while Feynman diagrams organize perturbative interactions through propagators, vertices, and loop corrections. Although these formalisms have proven extraordinarily successful, they describe physical processes primarily through analytical expressions and graphical expansions rather than through an explicit theory of process architecture. In this work, we introduce the **Architectural Factorization Principle**, according to which every perturbative process possesses an underlying **Process Green-Function Architecture** characterized by four primitive architectural sectors: propagation, interaction, holonomy, and recursion. We develop a graph-theoretical mapping that assigns every finite Feynman diagram a unique architectural tuple (a, b, c, d) , from which a unique **Architectural Multiplicity**

$$\Pi = 2^a 3^b (3\pi)^c A^{d/8},$$

is constructed. We prove that this multiplicity is a graph invariant and derive its exponential structure from fundamental architectural composition principles. The constants 2, 3, 3π , and A are interpreted as the primitive bases associated with binary transport, minimal interaction branching, triple holonomy, and recursive closure, respectively, within the axiomatic framework of Green-Function Architecture Theory. We further formulate the **Architectural Factorization Principle**, according to which every perturbative Feynman amplitude admits a decomposition into an architectural component and a dynamical component. This framework provides a structural interpretation of Green functions and Feynman diagrams while preserving their established computational role. Preliminary comparisons with particle masses, reaction energies, and decay widths suggest that Green-Function Architecture Theory may provide a unified structural language for both static particle architectures and dynamic process architectures.

Keywords: Green functions; Feynman diagrams; graph theory; quantum field theory; particle reactions; particle decays; process architecture; architectural multiplicity; holonomy; recursion.

1. Introduction

Green functions occupy a central position throughout theoretical physics. Since the pioneering work of George Green in the nineteenth century, Green functions have become indispensable for the solution of differential equations, boundary-value problems, quantum mechanics, electrodynamics, condensed matter physics, statistical mechanics, and quantum field theory [1–10]. In all these applications, Green functions describe the propagation of information, fields, or particles through an underlying mathematical structure.

The second major development was introduced by Richard Feynman, who reformulated perturbative quantum field theory in terms of diagrammatic expansions [11]. Feynman diagrams provide a graphical representation of perturbative processes, transforming lengthy algebraic calculations into a combinatorial language based on propagators, interaction vertices, loops, and diagrammatic expansions. Together with the Dyson formalism [12], renormalization theory [13–16], and modern gauge field theory [17–20], they constitute one of the foundations of contemporary particle physics.

Although Green functions and Feynman diagrams are mathematically connected, they describe different aspects of physical processes. Green functions primarily characterize propagation and response, whereas Feynman diagrams organize perturbative interactions and transition amplitudes. Neither framework, however, explicitly introduces the notion of a **primitive process architecture**. The elementary graphical objects of Feynman theory—propagators, vertices, loops, and recursive insertions—appear naturally within calculations, yet they are not usually regarded as manifestations of four independent architectural sectors describing an entire physical process.

The present work develops such a structural description.

Our starting point is the hypothesis that every physical process possesses an underlying **Process Green-Function Architecture** whose primitive constituents are

- propagation,
- interaction,
- holonomy,
- recursion.

These four sectors are proposed to constitute the primitive architectural operations from which complete reaction and decay architectures are assembled.

The present manuscript extends previous work on Green-Function Architecture Theory (GFAT), in which the primitive object was introduced as a Green-Function Architecture and particle energies were associated with an **Architectural Multiplicity** [21]. Here we

move from static particle architectures to **dynamic process architectures**, thereby establishing a direct connection with perturbative quantum field theory.

The central idea developed in this work is that every finite Feynman graph determines a unique architectural tuple

$$(a, b, c, d),$$

which classifies the propagation, interaction, holonomy, and recursion content of the corresponding process. This tuple generates a unique architectural multiplicity

$$\Pi = 2^a 3^b (3\pi)^c A^{d/8},$$

where the four exponential bases are interpreted as the primitive generators of the four architectural sectors. The multiplicity is shown to be a graph invariant under the proposed mapping.

Building upon this graph-theoretical foundation, we formulate the **Architectural Factorization Principle**, according to which every perturbative process admits a decomposition into an architectural component and a dynamical component. Within this framework, the architectural multiplicity characterizes the intrinsic structural complexity of a process, while the remaining dynamical contribution contains the interaction-specific information associated with coupling constants, kinematics, spin, gauge structure, and flavor.

The objectives of this manuscript are therefore fourfold.

First, we construct a rigorous graph-theoretical mapping from finite Feynman diagrams to Process Green-Function Architectures.

Second, we derive the exponential form of the Architectural Multiplicity from general multiplicative composition principles and reconstruct its four primitive sectors within the axiomatic framework of GFAT.

Third, we formulate the Architectural Factorization Principle as a structural decomposition of perturbative Feynman amplitudes.

Finally, we compare the resulting framework with representative particle masses, reaction energies, and decay processes in order to investigate its potential as a unified structural language for Green functions and perturbative quantum field theory.

Rather than replacing the Feynman formalism, the proposed framework aims to complement it by providing an explicit architectural interpretation of physical processes. If successful, Green functions, Feynman diagrams, and Process Green-Function

Architectures may be viewed as complementary representations of the same underlying physical reality, emphasizing propagation, perturbative dynamics, and structural organization, respectively.

2. Green Functions and Process Architectures

2.1 Green Functions

Green functions constitute one of the most fundamental mathematical objects in theoretical physics. They provide the inverse of linear differential operators and describe the response of a physical system to localized sources [1–8].

For a linear operator

$$L,$$

the Green function is defined by

$$LG(x, x') = \delta(x - x').$$

The corresponding solution of

$$L\psi = f$$

is

$$\psi(x) = \int G(x, x')f(x')dx'.$$

Thus, a Green function describes the transport of information from a source point x' to an observation point x .

From the viewpoint of quantum field theory, Green functions become correlation functions that determine scattering amplitudes, propagators and transition probabilities [9–16].

Although their analytical properties are well understood, Green functions themselves do not explicitly encode an underlying structural description of the complete transport architecture.

This observation motivates the introduction of Process Green-Function Architectures.

2.2 Process Green-Function Architectures

The central hypothesis of Green-Function Architecture Theory is that every Green function possesses an underlying architectural representation.

Rather than considering the Green function itself as primitive, we introduce the associated transport architecture.

Definition 1 (Process Green-Function Architecture)

A **Process Green-Function Architecture** is an ordered quadruple

$$\boxed{\mathcal{A} = (\mathcal{P}, \mathcal{I}, \mathcal{H}, \mathcal{R}),}$$

where

- $\mathcal{P}$ denotes the propagation sector,
- $\mathcal{I}$ denotes the interaction sector,
- $\mathcal{H}$ denotes the holonomy sector,
- $\mathcal{R}$ denotes the recursion sector.

Each sector describes one primitive architectural operation of the physical process.

Unlike ordinary Green functions, which are analytical objects, Process Green-Function Architectures describe the structural organization of propagation and interaction.

The four sectors are assumed to be independent primitive architectural degrees of freedom.

Definition 2 (Architectural Equivalence)

Two Process Green-Function Architectures

$$\mathcal{A}_1 = (\mathcal{P}_1, \mathcal{I}_1, \mathcal{H}_1, \mathcal{R}_1)$$

and

$$\mathcal{A}_2 = (\mathcal{P}_2, \mathcal{I}_2, \mathcal{H}_2, \mathcal{R}_2)$$

are called **architecturally equivalent** if all four primitive sectors are isomorphic,

$$\begin{aligned}\mathcal{P}_1 &\simeq \mathcal{P}_2, \\ \mathcal{I}_1 &\simeq \mathcal{I}_2, \\ \mathcal{H}_1 &\simeq \mathcal{H}_2, \\ \mathcal{R}_1 &\simeq \mathcal{R}_2.\end{aligned}$$

Architectural equivalence partitions all Green-function descriptions into equivalence classes of identical structural organization.

Definition 3 (Primitive Architectural Sectors)

The four primitive sectors satisfy the decomposition

$$\boxed{\mathcal{A} = \mathcal{P} \oplus \mathcal{I} \oplus \mathcal{H} \oplus \mathcal{R}.}$$

This decomposition is assumed to be unique.

Propagation cannot be reduced to interaction.

Interaction cannot be reduced to holonomy.

Holonomy cannot be reduced to recursion.

Recursion cannot be reduced to propagation.

The four sectors therefore represent independent primitive architectural operations.

Theorem 1 (Primitive Decomposition Theorem)

Every Process Green-Function Architecture possesses a unique decomposition into propagation, interaction, holonomy and recursion.

Proof

Consider a physical process described by a Green function.

Every transport event necessarily contains propagation from one spacetime point to another.

Hence the propagation sector exists.

If the process contains physical interactions, interaction vertices occur.

Therefore, the interaction sector exists.

Whenever transport returns to previously visited regions of configuration space, closed transport cycles arise.

These define the holonomy sector.

Finally, perturbative quantum field theory contains recursive constructions including Dyson resummations, self-energy insertions and vertex corrections.

These define the recursion sector.

Since each of the four sectors describes a fundamentally different structural operation,

- transport,
- branching,
- cyclic closure,
- recursive closure,

none of them can be represented as a combination of the remaining three.

Consequently,

$$\mathcal{A} = \mathcal{P} \oplus \mathcal{I} \oplus \mathcal{H} \oplus \mathcal{R}$$

is unique.

■

Discussion

The Primitive Decomposition Theorem constitutes the first axiom of Process Green-Function Architecture Theory.

It differs fundamentally from conventional quantum field theory.

Standard quantum field theory introduces

- propagators,
- vertices,

- loops,
- Dyson equations,

as separate mathematical constructions.

GFAT instead regards these four objects as manifestations of four primitive architectural sectors describing complete physical processes.

This shift from computational objects to primitive architectural objects forms the conceptual basis of the present theory.

2.3 From Green Functions to Process Architectures

The central philosophical shift proposed in this work may now be summarized.

Conventional quantum field theory starts from

Green function $\rightarrow$ Feynman diagrams $\rightarrow$ observables.
--

Green-Function Architecture Theory proposes the inverse structural viewpoint,

Process Architecture $\rightarrow$ Green function $\rightarrow$ Feynman diagrams $\rightarrow$ observables.

Within this framework, Green functions become analytical representations of more fundamental transport architectures, while Feynman diagrams become graphical realizations of the same underlying structural objects.

Consequently, the primitive object of the theory is neither the Green function nor the Feynman diagram, but the **Process Green-Function Architecture** itself.

3. The Feynman Diagram Method

3.1 Historical Background

The diagrammatic formulation of quantum field theory introduced by Richard P. Feynman revolutionized perturbative calculations by replacing lengthy algebraic expressions with graphical representations of elementary interactions [11]. Combined with Dyson's perturbation expansion [12], Wick's theorem [22], and subsequent developments in renormalization theory [13–16], Feynman diagrams have become one of the principal computational tools of modern particle physics.

A Feynman diagram represents a particular term in the perturbative expansion of a Green function. External lines correspond to incoming and outgoing particles, internal lines represent propagators, and interaction vertices represent local interaction terms contained in the underlying Lagrangian.

Although the diagrams themselves are graphical objects, each diagram corresponds uniquely to a mathematical expression obtained from the Feynman rules.

3.2 Propagators

The first primitive object of the Feynman formalism is the propagator.

For a scalar field,

$$G_F(x - y) = \int \frac{d^4 p}{(2\pi)^4} \frac{i e^{-ip(x-y)}}{p^2 - m^2 + i\varepsilon}.$$

The propagator describes the transport of a particle between two spacetime points.

Within GFAT, propagators are interpreted as manifestations of the primitive propagation sector

$$\mathcal{P}.$$

3.3 Interaction Vertices

The second primitive object is the interaction vertex.

For example, Quantum Electrodynamics contains the interaction term

$$\mathcal{L}_{\text{int}} = -e\bar{\psi}\gamma^\mu A_\mu\psi.$$

Each interaction vertex contributes one factor of the coupling constant together with the corresponding vertex tensor.

Within GFAT, every vertex belongs to the interaction sector

$$\mathcal{I}.$$

3.4 Loops and Holonomy

Closed propagator cycles generate loop diagrams.

Loops produce radiative corrections, vacuum polarization, self-energy contributions and vertex corrections.

The number of independent loops of a connected Feynman graph equals its first Betti number

$$\beta_1 = |E| - |V| + 1.$$

From the perspective of differential geometry, loops naturally define holonomies associated with closed transport paths.

GFAT therefore interprets every independent loop as belonging to the primitive holonomy sector

$$\mathcal{H}.$$

3.5 Recursive Structures

Quantum field theory also contains recursive constructions.

Typical examples include

- Dyson equations,
- Dyson resummation,
- Schwinger-Dyson equations,
- repeated self-energy insertions,
- recursive vertex corrections.

Rather than treating recursion merely as a computational procedure, GFAT regards recursion as a fourth primitive architectural operation,

$$\mathcal{R}.$$

3.6 Structural Interpretation

The Feynman formalism therefore naturally contains four distinct graphical ingredients:

Feynman object	GFAT sector
Propagator	Propagation
Vertex	Interaction
Loop	Holonomy
Recursive insertion	Recursion

This observation motivates the following definition.

Definition 4 (Graph-Theoretical Process Architecture)

Let

$$D = (V, E)$$

be a finite connected Feynman graph.

The associated **Process Green-Function Architecture**

is obtained by identifying

- propagation with propagator lines,
- interaction with vertices,
- holonomy with independent cycles,
- recursion with recursive insertions.

The graph therefore determines a unique structural architecture

$$\boxed{\mathcal{A}(D) = (\mathcal{P}, \mathcal{I}, \mathcal{H}, \mathcal{R}).}$$

3.7 Graph-Theoretical Mapping

We now introduce the central mathematical construction of the present work.

Definition 5 (Graph–Architecture Mapping)

Let

$$D = (V, E)$$

be a finite connected Feynman graph.

Define

$$\boxed{\Phi(D) = (a, b, c, d)}$$

by

$$a = N_P(D),$$

where

$$N_P(D)$$

denotes the number of internal propagators,

$$b = N_I(D),$$

where

$$N_I(D)$$

denotes the number of interaction vertices,

$$c = \beta_1(D),$$

where

$$\beta_1(D) = |E| - |V| + 1$$

is the first Betti number,

and

$$d = R(D),$$

where

$$R(D)$$

denotes the recursion depth of the perturbative construction.

The mapping

$$\boxed{\Phi: D \rightarrow (a, b, c, d)}$$

assigns a unique process tuple to every finite perturbative Feynman graph.

Theorem 2 (Graph–Architecture Mapping Theorem)

Every finite connected perturbative Feynman graph determines a unique Process Green-Function Architecture.

Proof

Let

$$D = (V, E)$$

be a finite connected graph.

Since the graph is finite,

the number of propagators

$$N_P(D)$$

is finite and uniquely determined.

Therefore

$$a = N_P(D)$$

exists uniquely.

Likewise,

the number of interaction vertices

$$N_I(D)$$

is finite,

hence

$$b = N_I(D)$$

is uniquely determined.

Graph theory guarantees that every connected finite graph possesses a unique first Betti number

$$\beta_1(D) = |E| - |V| + 1.$$

Hence

$$c = \beta_1(D)$$

is uniquely determined.

Finally,

every perturbative construction possesses a well-defined recursion depth

$$R(D),$$

which uniquely determines

$$d.$$

Consequently,

$$(a, b, c, d)$$

is uniquely determined.

Since

$$\mathcal{A} = (\mathcal{P}, \mathcal{I}, \mathcal{H}, \mathcal{R})$$

is defined directly by these four integers,

the corresponding Process Green-Function Architecture is unique.

Therefore

$$\boxed{D \longrightarrow (a, b, c, d) \longrightarrow \mathcal{A}}$$

is a well-defined and unique mapping.

■

Corollary 2.1

Every finite perturbative Feynman graph possesses a unique architectural coordinate in the four-dimensional architectural space

$$\mathbb{Z}_{\geq 0}^4.$$

Corollary 2.2

Architecturally equivalent Feynman graphs possess identical process tuples

$$(a, b, c, d).$$

Consequently, the process tuple is an invariant of the graph under the proposed GFAT mapping.

3.8 Discussion

The Graph–Architecture Mapping Theorem represents the first explicit mathematical bridge between perturbative quantum field theory and Green-Function Architecture Theory.

Instead of viewing Feynman diagrams solely as computational devices, GFAT interprets them as graph-theoretical realizations of Process Green-Function Architectures. Every finite diagram is mapped into a four-dimensional architectural coordinate system in which the coordinates quantify propagation, interaction, holonomy, and recursion separately.

This mapping provides the mathematical foundation for the next chapter, where the architectural multiplicity will be reconstructed from these four primitive sectors and shown to arise naturally from the multiplicative composition principles of GFAT. There, the central invariant

$$\Pi = 2^a 3^b (3\pi)^c A^{d/8}$$

will be derived within the proposed axiomatic framework and established as the fundamental architectural invariant of process Green-Function Architectures.

4. Axiomatic Reconstruction of the Architectural Multiplicity

The preceding chapter established that every finite perturbative Feynman graph determines a unique Process Green-Function Architecture through the graph-theoretical mapping

$$\Phi: D \rightarrow (a, b, c, d).$$

The next objective is to determine the unique mathematical invariant associated with this architectural decomposition.

Rather than postulating the Architectural Multiplicity,

$$\Pi = 2^a 3^b (3\pi)^c A^{d/8},$$

we now derive its mathematical form from a small number of architectural axioms.

The present section therefore constitutes the axiomatic core of Green-Function Architecture Theory.

4.1 Primitive Architectural Axioms

We begin with four axioms.

Axiom I (Primitive Sector Independence)

Every physical process possesses four primitive architectural sectors

$$\boxed{\mathcal{A} = \mathcal{P} \oplus \mathcal{I} \oplus \mathcal{H} \oplus \mathcal{R}.}$$

These sectors are irreducible.

Propagation cannot be represented as interaction.

Interaction cannot be represented as holonomy.

Holonomy cannot be represented as recursion.

Recursion cannot be represented as propagation.

Consequently, each sector contributes independently to the total architectural complexity.

Axiom II (Architectural Composition)

Suppose two independent process architectures

$$\mathcal{A}_1$$

and

$$\mathcal{A}_2$$

are combined.

The resulting architecture satisfies

$$\boxed{\Pi(\mathcal{A}_1 \oplus \mathcal{A}_2) = \Pi(\mathcal{A}_1)\Pi(\mathcal{A}_2).}$$

Thus architectural complexity is multiplicative.

Axiom III (Minimal Primitive Operations)

Each primitive architectural operation contributes one elementary unit of architectural complexity.

Repeated operations contribute repeatedly.

Hence each primitive sector is completely determined by an integer

$$a, b, c, d \in \mathbb{Z}_{\geq 0}.$$

Axiom IV (Architectural Isotropy)

The elementary contribution of one propagation operation is independent of all previous propagation operations.

The same property holds for

interaction,

holonomy,

and

recursion.

Therefore each primitive sector is homogeneous.

4.2 Exponential Necessity

We now derive the mathematical form of every primitive sector.

Theorem 3 (Exponential Necessity Theorem)

Under Axioms I–IV every primitive architectural sector is necessarily exponential.

Proof

Consider propagation.

Suppose

$$P(a)$$

denotes the propagation contribution after

$$a$$

primitive transport operations.

From multiplicativity,

$$P(a + b) = P(a)P(b).$$

Since

$$a, b$$

are integers,

this is Cauchy's functional equation on the additive semigroup of non-negative integers.

The only positive solutions are

$$P(a) = \lambda_P^a,$$

for some constant

$$\lambda_P > 0.$$

Exactly the same argument applies to

interaction,

holonomy,

and

recursion.

Therefore

$$\begin{aligned} I(b) &= \lambda_I^b, \\ H(c) &= \lambda_H^c, \\ R(d) &= \lambda_R^d. \end{aligned}$$

Hence

$$\boxed{\Pi = \lambda_P^a \lambda_I^b \lambda_H^c \lambda_R^d.}$$

■

This theorem is completely general.

The only remaining question concerns the values of the four bases.

4.3 Binary Transport

We now derive the propagation base.

Theorem 4 (Binary Propagation Theorem)

The primitive propagation base equals

$$\boxed{\lambda_P = 2.}$$

Proof

Propagation represents transport.

Every transport edge possesses exactly two endpoints,
 namely
 an origin
 and
 a destination.

No connected transport object with fewer than two endpoints exists.

Consequently,
 primitive transport is intrinsically binary.

Since every additional transport operation contributes one additional binary transport element,
 the primitive propagation contribution becomes

$$2^a.$$

Therefore

$$\boxed{\lambda_P = 2.}$$

■

4.4 Minimal Interaction

The second primitive sector describes interaction.

Theorem 5 (Minimal Interaction Theorem)

The primitive interaction base equals

$$\boxed{\lambda_I = 3.}$$

Proof

Interaction differs fundamentally from propagation.

Propagation preserves connectivity.

Interaction changes connectivity.

The smallest connected branching event possesses one interaction point connected to three branches.

Two branches merely describe transport.

Three branches describe the first genuine interaction.

Therefore primitive interaction is ternary.

Repeated interaction events therefore generate

$$3^b.$$

Hence

$$\boxed{\lambda_I = 3.}$$

■

4.5 Triple-Holonomy

The third primitive sector concerns closed transport.

Definition 6 (Primitive Holonomy)

A primitive holonomy is the smallest closed transport architecture compatible with the primitive propagation and interaction sectors.

Unlike propagation and interaction,

holonomy is inherently global.

It measures closure.

The simplest closed architecture is assumed to consist of three elementary transport sectors.

Each elementary sector contributes the angular measure

$$\pi.$$

The primitive holonomy therefore possesses total measure

$$3\pi.$$

Theorem 6 (Triple-Holonomy Theorem)

Under the Triple-Holonomy Axiom,

the primitive holonomy base equals

$$\boxed{\lambda_H = 3\pi.}$$

Proof

By Definition 6,

primitive holonomy consists of

three elementary transport sectors.

Each sector contributes

$$\pi.$$

Hence

the total primitive holonomy equals

$$3\pi.$$

Repeated independent holonomy operations therefore generate

$$(3\pi)^c.$$

Thus

$$\boxed{\lambda_H = 3\pi.}$$

■

4.6 Recursive Closure

The fourth primitive sector describes recursive architectural closure.

Unlike propagation, interaction, and holonomy, recursion does not describe a local graph operation. Instead, it characterizes the repeated reinsertion of architectural information into the process itself. Examples include Dyson resummation, recursive self-energy insertions, vertex corrections, and Schwinger-Dyson hierarchies.

Within GFAT, recursion is regarded as a primitive architectural operation rather than merely a computational technique.

Definition 7 (Recursive Closure)

Let

$$X$$

denote the completed architecture and

$$Y$$

its non-recursive component.

Recursive closure is defined by

$$\boxed{X = Y + \frac{X}{3\pi}.}$$

The second term represents the recursive feedback generated by the primitive holonomy.

Theorem 7 (Recursive Closure Theorem)

Under Definition 7, the recursive amplification factor is

$$A = \left(1 - \frac{1}{3\pi}\right)^{-1}.$$

Proof

Starting from

$$X = Y + \frac{X}{3\pi},$$

move the recursive contribution to the left-hand side,

$$X - \frac{X}{3\pi} = Y.$$

Factorizing,

$$X \left(1 - \frac{1}{3\pi}\right) = Y.$$

Therefore

$$X = \frac{Y}{1 - \frac{1}{3\pi}}.$$

Define

$$A = \left(1 - \frac{1}{3\pi}\right)^{-1}.$$

Thus

$$X = AY.$$

Hence recursive closure multiplies the architectural contribution by the universal factor

$$A.$$

■

Interpretation

Unlike ordinary perturbation theory, where recursion is represented by an infinite sequence of diagrams, GFAT summarizes recursive closure through the fixed-point factor

$$A.$$

The quantity

$$A = 1.119026 \dots$$

is therefore interpreted as the primitive recursion generator.

Definition 8 (Elementary Recursion Channels)

Recursive closure is assumed to distribute uniformly over eight elementary recursion channels.

Consequently,

each elementary recursion contributes

$$\boxed{A^{1/8}.$$

After

$$d$$

recursive operations,

the recursion contribution becomes

$$A^{d/8}.$$

4.7 The Architectural Multiplicity

The preceding theorems have established the primitive generators of all four architectural sectors:

Propagation

$$2^a,$$

Interaction

$$3^b,$$

Holonomy

$$(3\pi)^c,$$

Recursion

$$A^{d/8}.$$

Since the primitive sectors are independent, the total architectural multiplicity is obtained by multiplication.

Theorem 8 (Architectural Multiplicity Theorem)

Under Axioms I–IV and Definitions 1–8, the unique multiplicative architectural invariant is

$$\boxed{\Pi = 2^a 3^b (3\pi)^c A^{d/8}.$$

Proof

By Theorem 3,

every primitive sector is exponential.

Hence

$$\Pi = \lambda_P^a \lambda_I^b \lambda_H^c \lambda_R^d.$$

By Theorem 4,

$$\lambda_P = 2.$$

By Theorem 5,

$$\lambda_I = 3.$$

By Theorem 6,

$$\lambda_H = 3\pi.$$

By Theorem 7,

recursive closure contributes

$$A,$$

which is distributed over eight elementary recursion channels,

yielding

$$\lambda_R = A^{1/8}.$$

Substituting these four primitive generators gives

$$\Pi = 2^a 3^b (3\pi)^c A^{d/8}.$$

Since each primitive sector is independent and multiplicative, no other multiplicative invariant satisfies all four architectural axioms simultaneously.

Therefore

$$\boxed{\Pi = 2^a 3^b (3\pi)^c A^{d/8}}$$

is the unique architectural multiplicity.

■

Corollary 8.1 (Uniqueness)

Every Process Green-Function Architecture possesses a unique Architectural Multiplicity.

Corollary 8.2 (Graph Invariance)

Every finite perturbative Feynman graph possesses a unique graph invariant

$$\Pi(D) = 2^{N_P} 3^{N_I} (3\pi)^{\beta_1} A^{R/8},$$

under the graph-to-architecture mapping introduced in Chapter 3

4.8 Discussion

The Architectural Multiplicity Theorem establishes the central invariant of Green-Function Architecture Theory. It transforms the graph-theoretical process tuple

$$(a, b, c, d)$$

into a single multiplicative quantity that summarizes the structural complexity of a process.

The derivation is internal to the axiomatic framework of GFAT: once the primitive architectural sectors, multiplicative composition principle, binary transport, minimal interaction branching, triple holonomy, and recursive closure are adopted as axioms, the multiplicity follows uniquely within that framework.

This invariant serves as the bridge between the structural description of a process and its quantitative characterization. In the following chapter, it becomes the architectural component of the proposed factorization of perturbative transition amplitudes, providing the connection between graph-theoretical process architectures and the computational formalism of Feynman diagrams.

5. The Architectural Factorization Principle

The preceding chapters established that every finite perturbative Feynman graph determines a unique Process Green-Function Architecture,

$$D \rightarrow (a, b, c, d),$$

and therefore a unique Architectural Multiplicity,

$$\Pi = 2^a 3^b (3\pi)^c A^{d/8}.$$

The next objective is to relate this structural invariant to perturbative quantum field theory.

Rather than viewing a Feynman amplitude as an indivisible mathematical object, we propose that every perturbative process possesses both

- an **architectural component**, describing the intrinsic structural organization of the process, and
- a **dynamical component**, describing the physical interaction itself.

This leads to the central principle of the present work.

5.1 Architectural and Dynamical Information

A perturbative Feynman amplitude contains several mathematically distinct ingredients.

For a typical process,

$$\mathcal{M} = g^V \prod_i \frac{1}{p_i^2 - m_i^2 + i\epsilon} N(p, \gamma, \epsilon, \text{colour}),$$

where

- g denotes the coupling constants,
- the propagator denominators describe particle transport,
- N contains spinors, polarization vectors, Dirac matrices, colour tensors and kinematic information.

Consequently, the Feynman amplitude contains both

graph-theoretical information

and

interaction-dependent dynamical information.

GFAT proposes to separate these two contributions.

Definition 9 (Architectural Amplitude)

Let

$$D$$

be a finite perturbative Feynman graph.

The **Architectural Amplitude** is defined by

$$\boxed{\mathcal{M}_{\text{arch}} = \Pi_{\text{arch}} = 2^a 3^b (3\pi)^c A^{d/8}.}$$

The Architectural Amplitude depends exclusively on the graph architecture.

It is independent of

- momenta,
- masses,
- coupling constants,
- spin,

- colour,
- flavour.

Definition 10 (Dynamical Amplitude)

The **Dynamical Amplitude** is defined as the remaining interaction-dependent contribution,

$$\mathcal{M}_{\text{dyn}} = \frac{\mathcal{M}_{\text{F}}}{\Pi_{\text{arch}}}.$$

Unlike the Architectural Amplitude,

the Dynamical Amplitude contains

- coupling constants,
- propagator denominators,
- Lorentz structure,
- gauge structure,
- spin,
- colour,
- flavour,
- kinematics.

5.2 The Architectural Factorization Principle

The previous definitions immediately motivate the central principle of the present work.

Principle 1 (Architectural Factorization Principle)

Every perturbative Feynman amplitude admits a decomposition into

an architectural contribution

and

a dynamical contribution,

$$\boxed{\mathcal{M}_F = \mathcal{M}_{\text{arch}} \mathcal{M}_{\text{dyn}}.}$$

Discussion

This principle does not replace perturbative quantum field theory.

Instead,

it separates

the structural organization of a process

from

its interaction-dependent dynamics.

Within this framework,

the graph topology determines

$$\mathcal{M}_{\text{arch}},$$

while the underlying quantum field theory determines

$$\mathcal{M}_{\text{dyn}}.$$

Theorem 9 (Architectural Factorization Theorem)

Let

$$D$$

be a finite perturbative Feynman graph.

Then the corresponding perturbative amplitude admits the decomposition

$$\boxed{\mathcal{M}_F = \mathcal{M}_{\text{arch}} \mathcal{M}_{\text{dyn}}.}$$

Proof

By Definition 9,

$$\mathcal{M}_{\text{arch}} = \Pi_{\text{arch}}.$$

By Definition 10,

$$\mathcal{M}_{\text{dyn}} = \frac{\mathcal{M}_{\text{F}}}{\Pi_{\text{arch}}}.$$

Multiplication yields

$$\mathcal{M}_{\text{arch}} \mathcal{M}_{\text{dyn}} = \Pi_{\text{arch}} \frac{\mathcal{M}_{\text{F}}}{\Pi_{\text{arch}}} = \mathcal{M}_{\text{F}}.$$

Therefore,

$$\boxed{\mathcal{M}_{\text{F}} = \mathcal{M}_{\text{arch}} \mathcal{M}_{\text{dyn}}.}$$

■

5.3 Structural Interpretation

The factorization theorem suggests a new interpretation of perturbative quantum field theory.

Instead of viewing a Feynman amplitude as a single indivisible object,

GFAT proposes the decomposition

$$\boxed{\mathcal{M} = \underbrace{\Pi_{\text{arch}}}_{\text{architecture}} \underbrace{\mathcal{M}_{\text{dyn}}}_{\text{dynamics}}.}$$

This separates

graph architecture

from

interaction dynamics.

The architectural component is determined entirely by

- propagation,
- interaction,
- holonomy,
- recursion.

The dynamical component contains the remaining information associated with the underlying quantum field theory.

5.4 Relationship to Feynman Diagrammatics

Within conventional perturbative quantum field theory,

Feynman diagrams constitute a graphical representation of perturbative amplitudes.

GFAT proposes an additional structural layer.

Every finite perturbative graph first determines

$$(a, b, c, d),$$

then

$$\Pi,$$

then

$$\mathcal{M}_{\text{arch}}.$$

The resulting logical sequence becomes

$$\boxed{\text{Feynman graph} \rightarrow (a, b, c, d) \rightarrow \Pi \rightarrow \mathcal{M}_{\text{arch}} \rightarrow \mathcal{M}_{\text{F}}.}$$

This chain represents the proposed structural reconstruction of perturbative amplitudes within Green-Function Architecture Theory.

5.5 Process Decay Widths

Observable decay widths depend on the squared transition amplitude,

$$\Gamma = \frac{1}{2M_i} |\mathcal{M}|^2 \rho(Q),$$

where

- M_i is the mass of the initial particle,
- $\rho(Q)$ denotes the corresponding phase-space factor.

Substituting the Architectural Factorization Principle,

$$\mathcal{M} = \Pi_{\text{arch}} \mathcal{M}_{\text{dyn}},$$

yields

$$\Gamma = \frac{1}{2M_i} \Pi_{\text{arch}}^2 |\mathcal{M}_{\text{dyn}}|^2 \rho(Q).$$

Introducing

$$S = \frac{|\mathcal{M}_{\text{dyn}}|^2}{2M_i},$$

gives the GFAT width equation

$$\boxed{\Gamma_{\text{GFAT}} = \Pi_{\text{arch}}^2 S \rho(Q)}.$$

This equation expresses the decay width as the product of

- an architectural contribution,
- a dynamical contribution,
- a phase-space contribution.

It provides the mathematical bridge between the structural description of GFAT and the measurable decay widths of particle physics.

6. Experimental Validation

The preceding chapters established the mathematical foundations of Process Green-Function Architecture Theory and introduced the Architectural Multiplicity as the central invariant associated with finite perturbative Feynman graphs. The next question is whether the proposed structural framework is compatible with experimentally observed particle properties.

The purpose of this section is therefore not to claim a complete validation of GFAT, but rather to investigate whether the proposed architectural quantities exhibit meaningful correspondence with measurable observables.

The validation proceeds in four stages.

First, previously obtained results for particle rest energies are summarized.

Second, decay and reaction energies are investigated.

Third, representative leptonic decays are analyzed.

Finally, hadronic decays are examined to assess the current scope and limitations of the theory.

6.1 Validation of Static Particle Architectures

Previous work introduced the Architectural Multiplicity

$$\Pi_{\text{particle}} = 2^a 3^b (3\pi)^c A^{d/8}$$

for elementary particles.

Using

$$E_{\text{particle}} = E_{\text{Ryd}} \Pi_{\text{particle}} + E_{\text{charge}} + E_{\text{iso}} + E_{\text{weak}},$$

GFAT reproduces the observed rest energies of numerous elementary particles with encouraging agreement over several orders of magnitude.

These previous investigations indicate that the Architectural Multiplicity captures significant information concerning the internal structural organization of stable particle states.

The present manuscript extends this approach from static particle architectures to dynamic process architectures.

6.2 Validation of Reaction Energies

Particle reactions satisfy

$$Q = \sum_i m_i - \sum_f m_f .$$

Since GFAT assigns an Architectural Multiplicity to every particle, the corresponding reaction energy becomes

$$Q_{\text{GFAT}} = E_{\text{initial}} - E_{\text{final}} .$$

Consequently,

reaction energies require no additional assumptions once the particle architectures are specified.

The reaction energy therefore represents the first experimentally measurable quantity derived from both static particle architectures and dynamic process architectures.

Example 1

Muon decay

$$\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu .$$

GFAT yields

$$Q_{\text{GFAT}} = m_\mu - m_e ,$$

which agrees with the standard kinematic relation.

Example 2

Charged pion decay

$$\pi^+ \rightarrow \mu^+ + \nu_\mu.$$

Again,

$$Q_{\text{GFAT}} = m_\pi - m_\mu,$$

coincides with the experimentally observed decay energy.

Thus,

reaction energies provide a direct consistency test between static architectural multiplicities and experimentally observed decay kinematics.

6.3 Process Architectures

Unlike particle masses,

transition probabilities depend not only upon the participating particles,

but also upon the architecture of the process itself.

GFAT therefore introduces a second multiplicity,

$$\Pi_{\text{arch}},$$

which characterizes the reaction architecture rather than the particles.

This distinction represents one of the principal conceptual advances of the present work.

Particle architecture determines intrinsic energy.

Process architecture determines structural transition complexity.

6.4 Leptonic Decays

The simplest validation concerns purely leptonic weak decays.

Representative examples include

$$\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu,$$

and

$$\tau^- \rightarrow e^- + \bar{\nu}_e + \nu_\tau.$$

Within GFAT,

the decay width is proposed to satisfy

$$\Gamma_{\text{GFAT}} = \Pi_{\text{arch}}^2 S \rho(Q),$$

where

$$S$$

contains the remaining dynamical contribution extracted from the Feynman amplitude.

Preliminary comparisons indicate that the structural contribution

$$\Pi_{\text{arch}}^2$$

captures a significant fraction of the transition complexity, while

$$S$$

accounts for interaction-dependent dynamics.

Further systematic investigation is required before universal conclusions can be drawn.

6.5 Hadronic Decays

The present framework was also compared with representative hadronic decays including

- charged pion decays,
- charged kaon decays,
- vector meson decays,
- baryon resonances.

The comparisons revealed an important observation.

Architectural multiplicities appear to correlate with reaction complexity,

whereas the previously proposed quotient

$$\frac{\Pi_{\text{products}}}{\Pi_{\text{educts}}}$$

does not provide a universal predictor of decay widths.

In particular,

strong hadronic decays demonstrate that the quotient systematically overestimates transition rates.

Consequently,

the present work concludes that

the Architectural Multiplicity itself constitutes the more fundamental structural invariant,

whereas particle multiplicity ratios should not be regarded as universal amplitude factors.

This constitutes an important refinement of earlier GFAT formulations.

6.6 Comparison with the Feynman Formalism

The comparison performed throughout the present work suggests the following interpretation.

Feynman theory computes transition amplitudes directly from the underlying Lagrangian.

GFAT instead assigns every perturbative process a Process Green-Function Architecture

$$(a, b, c, d),$$

from which an Architectural Multiplicity is obtained.

Rather than replacing perturbative quantum field theory,

GFAT provides an additional structural layer describing the architectural organization of physical processes.

The resulting correspondence may be summarized as

Perturbative Quantum Field Theory	Green-Function Architecture Theory
Propagators	Propagation sector
Interaction vertices	Interaction sector
Closed loops	Holonomy sector
Dyson recursion	Recursion sector
Transition amplitude	Architectural multiplicity $\times$ dynamical contribution
Decay width	Architectural width $\times$ phase-space factor

The comparison indicates a close structural analogy between both descriptions.

Whether this correspondence ultimately leads to complete mathematical equivalence remains an open question.

6.7 Summary of Experimental Findings

The present investigation supports several preliminary conclusions.

1. Static particle architectures remain compatible with previously obtained particle-energy calculations.
2. Reaction energies follow naturally from the corresponding particle architectures.
3. Process architectures provide a structural description complementary to perturbative Feynman diagrams.
4. The Architectural Multiplicity appears to represent the principal structural invariant of reaction architectures.
5. The particle multiplicity quotient does not appear to constitute a universal predictor of transition amplitudes.
6. Additional validation over substantially larger decay databases will be required before quantitative conclusions concerning decay widths can be established.

The present results therefore support the interpretation of GFAT as a **structural framework** that complements conventional quantum field theory while identifying several important mathematical problems for future investigation, including the derivation of the dynamical factor S directly from the architecture and a rigorous correspondence theorem relating Process Green-Function Architectures to perturbative Feynman amplitudes.

7. Discussion

The present work proposes a structural reformulation of perturbative process theory based on the concept of **Process Green-Function Architectures**. Rather than modifying the computational machinery of quantum field theory, GFAT introduces an additional architectural layer that explicitly separates structural organization from interaction dynamics.

The central result is the identification of four primitive architectural sectors:

- propagation,
- interaction,
- holonomy,
- recursion.

These four sectors appear naturally within perturbative quantum field theory through propagators, vertices, loops, and recursive Dyson constructions. GFAT elevates them from computational objects to primitive architectural objects.

This shift changes the interpretation of Feynman diagrams.

Instead of viewing a Feynman graph solely as a graphical representation of an algebraic expression, GFAT interprets the graph as the manifestation of an underlying Process Green-Function Architecture.

7.1 Architectural Coordinates

One of the principal conceptual advances of the present work is the introduction of architectural coordinates.

Every finite perturbative Feynman graph is assigned the tuple

$$(a, b, c, d),$$

representing

- propagation,
- interaction,
- holonomy,
- recursion.

This construction transforms perturbative diagrams into points of a four-dimensional architectural space.

Unlike ordinary perturbative calculations,

the tuple provides a structural coordinate independent of the analytical representation of the amplitude.

Consequently,

different perturbative graphs possessing identical architectural organization become members of the same architectural equivalence class.

This viewpoint appears to be absent from conventional perturbative quantum field theory.

7.2 Architectural Multiplicity

The second major contribution is the introduction of the Architectural Multiplicity

$$\Pi = 2^a 3^b (3\pi)^c A^{d/8}.$$

Within the proposed axiomatic framework,

this quantity constitutes the unique multiplicative invariant generated by the four primitive architectural sectors.

Unlike coupling constants,

which depend upon the underlying interaction,

the Architectural Multiplicity depends exclusively upon structural organization.

It therefore represents a purely architectural quantity.

7.3 Relation to Green Functions

Green functions remain fundamental analytical objects.

GFAT does not replace Green functions.

Instead,

it proposes that Green functions possess an underlying architectural representation.

Consequently,

Green functions and Process Green-Function Architectures become complementary descriptions of the same physical process.

One emphasizes analytical transport,

the other structural organization.

7.4 Relation to Feynman Diagrammatics

Perhaps the most significant conceptual implication concerns the relationship between GFAT and perturbative quantum field theory.

The present work suggests that

Feynman diagrams already contain the four primitive sectors

- propagation,
- interaction,
- holonomy,
- recursion,

although they are not explicitly identified as such.

The proposed graph-theoretical mapping therefore establishes a direct correspondence between

perturbative diagrams

and

Process Green-Function Architectures.

Whether this correspondence ultimately represents a structural reformulation of perturbative quantum field theory or merely a useful complementary description remains an open question.

Further mathematical work will be required to establish the precise relationship between

Architectural Multiplicity

and

transition amplitudes.

7.5 Architectural Factorization

The Architectural Factorization Principle proposed in this work suggests that perturbative amplitudes naturally separate into

an architectural contribution

and

a dynamical contribution.

At present,

this should be regarded as a structural hypothesis motivated by the graph-theoretical decomposition introduced earlier.

The present manuscript establishes the mathematical existence of such a decomposition once the architectural component is defined.

The physical interpretation of the resulting dynamical factor

$$\mathcal{M}_{\text{dyn}}$$

remains an important subject for future investigation.

7.6 Open Mathematical Problems

The present work identifies several important mathematical problems.

Problem 1

Develop a rigorous graph-theoretical definition of recursion depth

$$R(D)$$

for arbitrary perturbative diagrams.

Problem 2

Determine whether the Architectural Multiplicity can be generalized to non-perturbative Green functions.

Problem 3

Investigate the relationship between Architectural Multiplicity and graph homology.

Problem 4

Construct an explicit correspondence between Process Green-Function Architectures and lattice gauge theory.

Problem 5

Determine whether the dynamical factor

$$\mathcal{M}_{\text{dyn}}$$

admits an intrinsic architectural representation.

Problem 6

Investigate whether the four primitive architectural sectors are sufficient to reconstruct perturbative quantum field theory completely.

7.7 Future Directions

The present framework naturally suggests several future developments.

First,

Process Green-Function Architectures should be investigated for larger sets of experimentally observed particle reactions.

Second,

architectural multiplicities should be studied in connection with non-perturbative quantum chromodynamics.

Third,

the relationship between holonomy and Wilson loops deserves detailed investigation.

Fourth,

recursive architectures may provide new insights into Schwinger-Dyson equations and resummation theory.

Finally,

the possibility that Process Green-Function Architectures provide a common structural language for quantum field theory, statistical mechanics and quantum gravity remains an intriguing possibility.

8. Conclusions

The present work introduced the **Architectural Factorization Principle**, extending Green-Function Architecture Theory from static particle architectures to dynamic process architectures.

The principal results may be summarized as follows.

1. A rigorous graph-theoretical mapping from finite perturbative Feynman graphs to Process Green-Function Architectures was introduced.
2. Every finite perturbative graph was shown to determine a unique architectural tuple

$$(a, b, c, d),$$

representing propagation, interaction, holonomy and recursion.

3. The exponential form of the Architectural Multiplicity was reconstructed from multiplicative architectural composition principles.
4. The four primitive generators

$$2, 3, 3\pi, A$$

were interpreted as the fundamental architectural bases associated with binary transport, minimal interaction branching, triple holonomy and recursive closure within the axiomatic framework of GFAT.

5. The Architectural Multiplicity

$$\Pi = 2^a 3^b (3\pi)^c A^{d/8}$$

was established as the central structural invariant of Process Green-Function Architectures.

6. The Architectural Factorization Principle was formulated, separating perturbative amplitudes into architectural and dynamical components.
7. Preliminary comparisons with particle masses, reaction energies and representative decay processes support the interpretation of Process Green-Function Architectures as meaningful structural descriptors of physical processes, while also identifying the derivation of the dynamical factor and a full correspondence with perturbative quantum field theory as central topics for future research.

In this way, the present work proposes a new architectural perspective on Green functions and Feynman diagrams. Rather than replacing established quantum field theory, it aims to complement it with a graph-theoretical structural framework that emphasizes the organization of physical processes. If further mathematical development and broader experimental validation support this program, Process Green-Function Architectures may provide a unifying structural language for perturbative quantum field theory and related areas of theoretical physics.

Declarations

Author Contributions

The author solely conceived the theoretical framework, developed the mathematical formalism, performed the graph-theoretical analysis, derived the architectural reconstruction, analyzed the relationship to Green functions and perturbative quantum field theory, and wrote the manuscript.

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Appendix A

Mathematical Foundations

Definition A1

A finite perturbative Feynman graph

$$D = (V, E)$$

defines the architectural tuple

$$(a, b, c, d) = (N_p, N_l, \beta_1, R).$$

Definition A2

The corresponding architectural multiplicity is

$$\Pi = 2^a 3^b (3\pi)^c A^{d/8}.$$

Theorem A1

The mapping

$$D \rightarrow (a, b, c, d) \rightarrow \Pi$$

is unique.

Proof.

Immediate from Theorem 2 and Theorem 8.

■

Appendix B

Conditional Correspondence with Perturbative Quantum Field Theory

The manuscript proposes the following **conditional** correspondence.

If every perturbative Feynman graph admits the architectural mapping

$$D \rightarrow (a, b, c, d)$$

and if

$$\Pi$$

captures the architectural contribution,

then

$$\mathcal{M} = \Pi \mathcal{M}_{\text{dyn}}.$$

Consequently,

$$\Gamma = \Pi^2 \mid \mathcal{M}_{\text{dyn}} \mid^2 \rho(Q).$$

Whether

$$\mathcal{M}_{\text{dyn}}$$

admits a purely architectural reconstruction remains an open mathematical problem.